



Filmon Misgane

Applied Mathematician

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Key Qualifications

1. Proficient in implementing algorithms using **Python** and managing projects on **GitHub**, enhancing team collaboration and increasing efficiency.
2. Proficient in numerical methods, such as the Finite Element Method (**FEM**), Finite Volume Methods and **Spline Methods**, which are applied in scientific computing to solve complex problems.
3. Experience testing software frameworks with **PyTest** and using **FEniCSx**, **NumPy**, **SymPy** and **Pandas**
4. Excellent analytical skills with the ability to solve complex problems and **visualize data**.
5. Disciplined professional with strong skills in collaborating with interdisciplinary teams and working independently.



Work Experience

Present
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04.2026

Summer Researcher *University of Oslo*

- Working on scientific machine learning methods for subsurface energy storage, with a focus on Physics-Informed Neural Networks (PINNs) and Neural Operators.
- Contributing to the development of machine learning models for PDE-based subsurface flow and transport problems.
- Supporting applications in H₂ and CO₂ storage, including model development, training, and evaluation.
- Using Python-based tools for scientific computing, machine learning, and analysis.

Present
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04.2026

Summer Intern *University of Oslo*

- Working on a unified FEM inversion framework for time-harmonic shear-wave elastography.
- Contributing to an end-to-end elastography software pipeline, from data ingestion and mesh generation to reconstruction and visualization.
- Supporting FEM-based inversion, data processing, and integration with an existing elastography toolbox.
- Collaborating on quantitative evaluation of reconstruction methods and software development for medical imaging applications.

Present
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01.2025

Substitute Teacher *Oslo Katedralskole (VGS)*

- Substitute teacher, teaching mathematics at all levels.



GitHub

<https://github.com/Filmon492>



Volunteer Work

I have been active as a volunteer social worker in the Student Council at the Department of Mathematics, where I was responsible for putting up posters and assisting with cleanup after events.



Skills

- Python
- C++
- MATLAB



Job References

Professor Ulrik Skre Fjordholm
Department of Mathematics, University of Oslo
E-mail: ulriksf@math.uio.no
Phone: (+47) 22 85 60 29
Relationship: Master's thesis supervisor



Languages

Norwegian	Fluent
English	Fluent

Present
↑
08.2022

Mentor
ENT3R, University of Oslo (UiO)

- Motivate high school students to study STEM subjects while creating engaging activities and assisting with their homework.

Present
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05.2022

Interpreter
Tolknett AS, Oslo

- Communicate information clearly and understandably to ensure effective interpretation.



Education

Present
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08.2024

MSc. Mathematics for Applications
University of Oslo (UiO)

- Specialized in Partial Differential Equations (PDEs).
- Master thesis: Numerical Methods for Nonlocal Conservation Laws.
- With guidance from my supervisor, I have designed and implemented **solvers** in Python for the first and second order numerical schemes.

06.2024
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08.2021

BSc. Mathematics and Computer Science
University of Oslo (UiO)